Research Notes on BLEU score

BLEU score

>> Section 1

Basic setup A basic, first attempt at defining the BLEU score would take two arguments: a candidate string y ^ {\displaystyle {\hat {y}}} and a list of reference strings (y (1) , . . . , y (N)) {\displaystyle (y^{(1)},...,y^{(N)})} . The idea is that B L E U (y ^ ; y (1) , . . . , y (N)) {\displaystyle BLEU({\hat {y}};y^{(1)},...,y^{(N)})} should be close to 1 when y ^ {\displaystyle {\hat {y}}} is similar to y (1) , . . . , y (N) {\displaystyle y^{(1)},...,y^{(N)}} , and close to 0 if not. As an analogy, the BLEU score is like a language teacher trying to score the quality of a student translation y ^ {\displaystyle {\hat {y}}} by checking how closely it follows the reference answers y (1) , . . . , y (N) {\displaystyle y^{(1)},...,y^{(N)}} . Since in natural language processing, one should evaluate a large set of candidate strings, one must generalize the BLEU score to the case where one has a list of M candidate strings (called a "corpus") (y ^ (1) , ■ , y ^ (M)) {\displaystyle ({\hat {y}}^{(1)},\cdots ,{\hat {y}}^{(M)})} , and for each candidate string y ^ (i) {\displaystyle {\hat {y}}^{(i)}} , a list of reference candidate strings S i := (y (i , 1) , . . . , y (i , N i)) {\displaystyle S_{i}:=(y^{(i,1)},...,y^{(i,N_{i})})} . Given any string y = y 1 y 2 ■ y K {\displaystyle y=y_{1}y_{2}\cdots y_{K}} , and any integer n ≥ 1 {\displaystyle n\geq 1} , we define the set of its n-grams to be G n (y) = { y 1 ■ y n , y 2 ■ y n + 1 , ■ , y K − n + 1 ■ y K } {\displaystyle G_{n}(y)=\{y_{1}\cdots y_{n},y_{2}\cdots y_{n+1},\cdots ,y_{K-n+1}\cdots y_{K}\}} Note that it is a set of unique elements, not a multiset allowing redundant elements, so that, for example, G 2 (a b a b) = { a b , b a } {\displaystyle G_{2}(abab)=\{ab,ba\}} . Given any two strings s , y {\displaystyle s,y} , define the substring count C (s , y) {\displaystyle C(s,y)} to be the number of appearances of s {\displaystyle s} as a substring of y {\displaystyle y} . For example, C (a b , a b c b a b) = 2 {\displaystyle C(ab,abcbab)=2} . Now, fix a candidate corpus S ^ := (y ^ (1) , ■ , y ^ (M)) {\displaystyle {\hat {S}}:=({\hat {y}}^{(1)},\cdots ,{\hat {y}}^{(M)})} , and reference candidate corpus S = (S 1 , ■ , S M) {\displaystyle S=(S_{1},\cdots ,S_{M})} , where each S i := (y (i , 1) , . . . , y (i , N i)) {\displaystyle S_{i}:=(y^{(i,1)},...,y^{(i,N_{i})})} .

>> Section 2

In practice, however, using individual words as the unit of comparison is not optimal. Instead, BLEU computes the same modified precision metric using n-grams. The length which has the "highest correlation with monolingual human judgements"[6] was found to be four. The unigram scores are found to account for the adequacy of the translation, how much information is retained. The longer n-gram scores account for the fluency of the translation, or to what extent it reads like "good English".

>> Section 3

BLEU (bilingual evaluation understudy) is an algorithm for evaluating the quality of text which has been machine-translated from one natural language to another. Quality is considered to be the correspondence between a machine's output and that of a human: "the closer a machine translation is to a professional human translation, the better it is" – this is the central idea behind BLEU.[1] Invented at IBM in 2001, BLEU was one of the first metrics to claim a high correlation with human judgements of quality,[2][3] and remains one of the most popular automated and inexpensive metrics. Scores are calculated for individual translated segments—generally sentences—by comparing them with a set of good quality reference translations. Those scores are then averaged over the whole corpus to reach an estimate of the translation's overall quality. Intelligibility or grammatical correctness are not taken into account.[4] BLEU's output is always a number between 0 and 1. This value indicates how similar the candidate text is to the reference texts, with values closer to 1 representing more similar texts. Few human translations will attain a score of 1, since this would indicate that the candidate is identical to one of the reference translations. For this reason, it is not necessary to attain a score of 1. Because there are more opportunities to match, adding additional reference translations will increase the BLEU score.[5]

>> Section 4

Final formula There is not a single definition of BLEU, but a whole family of them, parametrized by the weighting vector w := (w 1 , w 2 , ■) {\displaystyle w:=(w_{1},w_{2},\cdots)} . It is a probability distribution on { 1 , 2 , 3 , ■ } {\displaystyle \{1,2,3,\cdots \}} , that is, ∑ i = 1 ∞ w i = 1 {\displaystyle \sum _{i=1}^{\infty }w_{i}=1} , and ∀ i ∈ { 1 , 2 , 3 , ■ } , w i ∈ [0 , 1] {\displaystyle \forall i\in \{1,2,3,\cdots \},w_{i}\in [0,1]} . With a choice of w {\displaystyle w} , the BLEU score is B L E U w (S ^ ; S) := B P (S ^ ; S) · exp ■ (∑ n = 1 ∞ w n ln ■ p n (S ^ ; S)) {\displaystyle BLEU_{w}({\hat {S}};S):=BP({\hat {S}};S)\cdot \exp \left(\sum _{n=1}^{\infty }w_{n}\ln p_{n}({\hat {S}};S)\right)} In words, it is a weighted geometric mean of all the modified n-gram precisions, multiplied by the brevity penalty. We use the weighted geometric mean, rather than the weighted arithmetic mean, to strongly favor candidate corpuses that are simultaneously good according to multiple n-gram precisions. The most typical choice, the one recommended in the original paper, is w 1 = ■ = w 4 = 1 4 {\displaystyle w_{1}=\cdots =w_{4}={\frac {1}{4}}} .